

Engineering Design Report

Team Name: - Piston Heads

Team ID: - 20200879



ABSTRACT

Quad Torc is a National Level Quad Bike designing competition organised by ISNEE. The event aims in bringing together engineering students from all around India, and provides an environment to showcase their various skills and engineering knowledge in designing and developing a complete Quad Bike. We have developed our conceptual design of a Quad Bike by considering stability, comfort, fuel economy, safety, and cost as the prime factors. The designing was done in CATIA V5 and analysis in ANSYS.

INTRODUCTION

Team PISTON HEADS is a group of aspiring engineers, formed by students of Muthoot Institute of Technology and Science affiliated to APJ Abdul Kalam Technological University. We started off by exploring various quad bike designs, previous year reports and attending webinars that made us confident enough to forge ahead with the competition. The designing was carried out by understanding the significance of every part of a Quad Bike. By doing necessary calculations and applying engineering knowledge, we were able to complete an exceptional design.

TECHNICAL SPECIFICATIONS

Wheel Base	41 inches
Max Width	49.75 inch
Ground Clearance	7 inches
Dry Weight	180 kg
Gross weight	250 kg
Max speed	70 kmph

Transmission	
Engine	KTM DUKE 250
Displacement	248.8 cc
Type	Manual 6-speed
Drive	Chain drive

Tyres	
Front tire	24*8*R12
Rear tire	24*8*R12

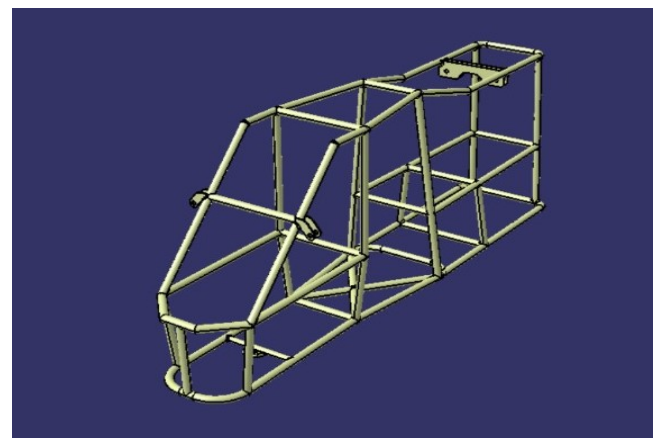
Steering	
Steering control	Manual
Mechanism	Tripod steering

Suspension	
Front suspension	Independent Dual A-arm
Rear suspension	H-arm

Brakes	
Front	Independent Hydraulic disc
Rear	Single Hydraulic disc

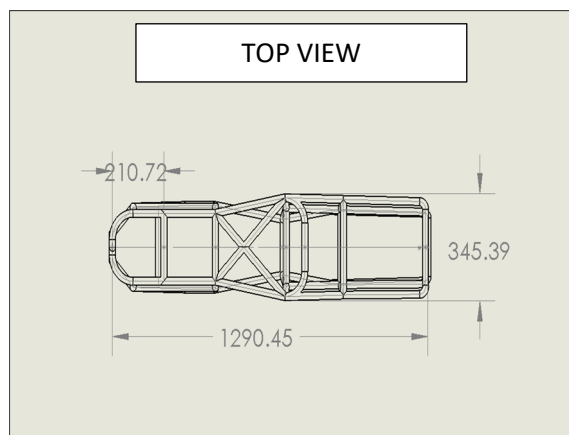
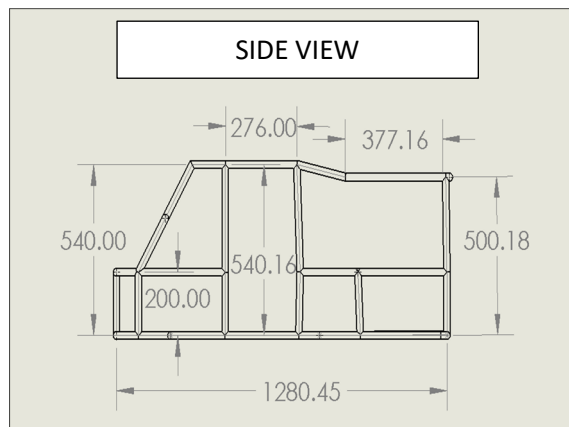
FRAME

The frame is designed so as to take all the impacts and also ensure comfort and stability of the vehicle. By understanding proper weight distribution of the vehicle, a compact and strong chassis was designed. The material used was AISI 4130 steel. The details of 4130 steel are as follows



isometric View of Frame

Material	AISI 4130 Steel
Yield Strength	460 MPa
Modulus of	205 GPa
Bending Stiffness	3635.1 N/m ²
Bending Strength	487 Nm
Density	7.85 g/cm ³
Melting Point	1432 °C
Poisons Ratio	0.27 – 0.30
Carbon	0.28 – 0.33 %
Molybdenum	0.15 – 0.25 %



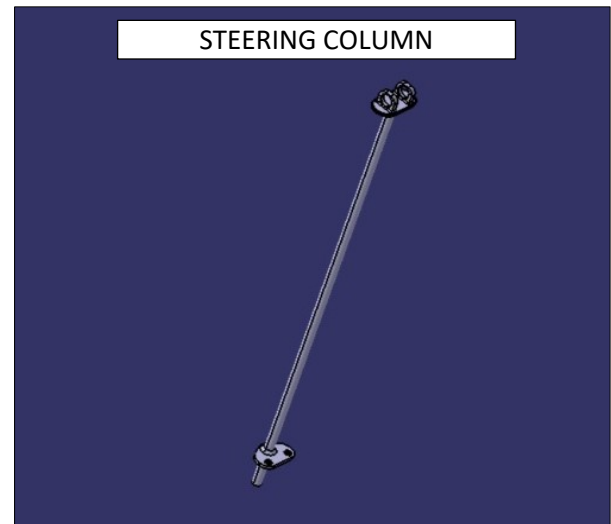
STEERING

Design Considerations: -

- The steering mechanism should be properly designed and aligned accordingly.
- It should be designed in such a manner that road shocks are not transmitted to the driver.
- The steering system must be such a way that it reduces effort from the driver.

The main parts of the steering system include:

- Handle Bar
- Steering Column
- Tripod Plate
- Tie Rod
- Tie Rod End
- Alignment Nut
- Knuckle



Parameters	Value
Wheel base	41 inches
Track Width	42 inches
Acceleration due to gravity	9.81m/s ²
Scrub Radius (r)	32.6 mm
King Pin Inclination	7.125
King Pin Offset	80 mm
Steering Arm length	70 mm
Front Axle Load	112.5 kg
Knuckle material	EN 18
Tie rod material	AISI 1020
Hub material	EN 18

BRAKING SYSTEM

A braking system is one of the most important components of a vehicle. They are required to stop the vehicle within the smallest possible distance. It has many components which are arranged in a way that it converts a vehicle's kinetic energy into heat energy which in turn stops the vehicle.

Design:

We have implemented an inboard braking system in the rear end and outboard in the front. Hydraulic disc brakes with independent hydraulic circuits have been used.

Calipers:

Floating calipers were used in the rear end and fixed calipers were used at the front. Floating calipers were selected due to their lightweight, availability and reliability. Using a fixed caliper at the front makes it compact and rigid, not only that but the stiffness of the caliper itself is greatly increased. This manifests itself in enhanced braking performance, pedal feel, and pad wear.

Master cylinder:

Single piston master cylinders were used with bore diameter 22 mm.

Calculations:

1.Kinetic Energy,

$$K. E = (m \times v^2) / 2$$

Value of V=40 km/h (average value)

$$V = 40 / 3.6$$

$$V = 11.11 \text{ m/s}$$

Where, m = mass of vehicle (180kg)
+ mass of driver (70 kg)

$$\begin{aligned} V &= \text{speed of the vehicle (m/s)} \\ &= (250 \times 11.11^2) / 2 \\ &= 15429.0125 \text{ J} \end{aligned}$$

2.Maximum Friction Force,

$$F = \mu \times m \times g$$

Where, μ = friction coefficient

m = mass of the vehicle

g = acceleration due to gravity

$$F = 0.7 \times 9.81 \times 250$$

$$= 1716.75 \text{ N}$$

3.Deceleration of the vehicle,

$$a = F / M$$

Where,

F = maximum friction force

M = mass of the vehicle

$$a = 1716.75 / 250$$

$$= 6.867 \text{ m/s}^2$$

4.Time taken to stop the vehicle,

$$t = V / a$$

$$= 11.11 / 6.867$$

$$= 1.618 \text{ s}$$

5.Total stopping distance =

$$V \times (\text{reaction time}) + (V^2 / 2\mu g)$$

(Reaction time for average humans is 1)

$$= 11.11 \times 1 + (11.11^2 / (2 \times 0.7 \times 9.81))$$

$$= 20.097 \text{ m}$$

6.Tangential braking force,

$$(B.F) = K.E / \text{Stopping Distance}$$

$$= 15429.0125 / 20.097$$

$$= 767.72 \text{ N}$$

7.Tangential force on each wheel,

$$F_t = B.F / 4$$

$$= 767.72 / 4$$

$$= 191.93 \text{ N}$$

8.Braking torque on wheel,

$$T_w = F_t \times R$$

$$= 767.72 \times (0.608 / 2)$$

$$= 233.38 \text{ Nm}$$

ENGINE AND TRANSMISSION

An engine or motor is a machine designed to convert one form of energy into mechanical energy.

Heat engines convert heat into work via various thermodynamic processes.

Engine generates a power of 30 PS at 9000 rpm and a torque of 24 Nm at 7500 rpm.

SPROCKET DESIGN:

NUMBER OF DRIVEN SPROCKET TEETH

Engine torque = 24 Nm

Number of teeth of front sprocket = 14

Torque on front sprocket(T1) = 24*10 = 240 Nm

No of teeth of rear sprocket = 46

The relation between teeth and torque is:

$$t_2 = (T_2 / T_1) * t_1$$

t1- No: of teeth of Driver sprocket

t2 - No: of teeth of Driven sprocket

T1- Torque on Driver Sprocket

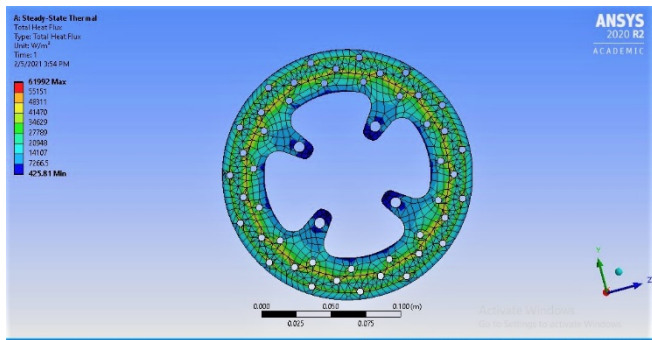
T2- Torque on Driven Sprocket

Which Gives,

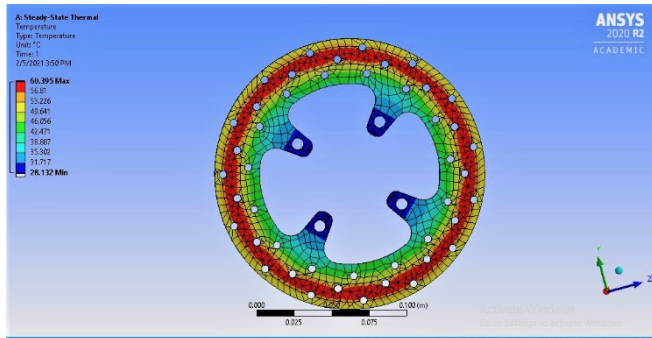
$$T_2 = (46/14) * 240$$

$$T_2 = 789 \text{ Nm}$$

Properties	Chain(inch)
Pitch	0.625
Roller diameter	0.4
Roller width	0.23
Sprocket thickness	0.31



heat flux of SS410 disk brake



temperature distribution of SS410 disk brake

Specifications	Dimensions	Description
Disc diameter front	180 mm	Fabricated
Disc diameter rear	200 mm	Fabricated
Tangential Brake force	233.38	For each wheel
Stopping distance	20.097	For 40 km/h
Kinetic energy	15429.012 J	
Time taken to stop the vehicle	1.618 s	
Deceleration of the vehicle	6.867 m/s ²	
Front calliper piston diameter	24 mm	
Rear caliper piston diameter	20 mm	
Material for disc	SS410	
Rear disc thickness	6 mm	
Front Disc Thickness	4 mm	

Sprocket reduction = 3.5 (Considering several dynamic constraints which can't be factored in theoretical calculations)

Speed of the vehicle = 60 Km/h

Chain drive calculation: -

In order to design, build and discuss chain drive systems it is necessary to understand the terminology and concepts associated with chain drive systems. Chains perform better than gears under shock loading conditions. Chain drives are a better choice for less

experienced builders working with a minimum of machine tools.

We have selected the chain no. 520 from YBN chart according to our requirement.

Properties of Chain selection are given below,

Calculation procedure: -

Transmission

A 6-gear manual transmission has been used with following gear ratios

1	2.66
2	1.857
3	1.421
4	1.142
5	0.956
6	0.84

Transmission calculation

Rolling force = F

$$F = \mu \times W$$

Where, μ = co-efficient of friction

$$\mu = 0.3 \text{ (assume)}$$

W = Total Weight = 250 kg

$$F = 735 \text{ N}$$

Rolling torque = $T_r = F \times R$

R = Average radius of front and rear wheels

$$= (r_1 + r_2) / 2$$

$$= 12 \text{ inches}$$

r = radius of front and rear wheel

$$T_r = 224.028 \text{ Nm.}$$

Net power = 11.5 KW at 7000 rpm

$$\text{Gear ratio} = 2.769 = i$$

$$Z_1 = \text{driving teeth} = 14$$

$$Z_2 = \text{driven teeth} = i \times z_1 = 46$$

$$\text{Ideal rpm} = 1420 \text{ rpm}$$

Kw rating of chain

$$= (\text{kw to be transmitted}) \times k_s / k_1 \times k_2$$

$$= (11.5 \times 1.7) / 1.5 \times 1.25$$

$$= 10.42 \text{ kw}$$

Ideal type of chain would be 10 B

$$\text{Pitch} = 15.875 \text{ mm}$$

Assumed centre to centre distance of sprocket =

$$= 30 \times \text{pitch}$$

$$= 476.25 \text{ mm} = a$$

$$R_1 = 149 (\text{pinion})$$

$$R_2 = 41$$

Length of chain =

$$= 3.14 (\text{radius of pinion} + \text{radius of sprocket}) + 2 \times c. d + (r \text{ of pinion} - R \text{ of sprocket})^2 / c. d$$

$$= 1556 \text{ mm}$$

No. of Chain link: - chain length / pitch

$$\text{Number of chain link} = 98.01 \approx 98 = L_n$$

According to chain design procedure we know that no. of chain link should be always even

SUSPENSION

Design methodology

Firstly, before designing the suspension system some basic parameters of the vehicle must be noted down.

They are:

- Wheel base: 1242.4mm
- Track width-
Front: 1043mm
Rear: 1043mm
- Tire radius: 304mm
- Tire width: 180mm
- Sprung mass: 130kg
- Unsprung mass: 100kg

These parameters are necessary for the designing the suspension system. We have chosen the Lotus Suspension Analysis (LSA) software for the designing process.

Lotus Suspension Analysis

It is a tool that will help you to design and analyse the vehicle suspension hardpoints to achieve the required suspension characteristics. Lotus Suspension Analysis is used for the initial layout of the vehicle. 3D models can be created and modified using LSA. The required parameters for the suspension design can be found out graphically and numerically using LSA. Suspensions can be articulated in bump/rebound, roll, steering and combined bump and steering (3D only) modes. Any number of results (camber angle, toe angle etc.) can be displayed graphically against bump motion, steering motion or roll motion. Several parameters are considered to get the hardpoints of the suspension system like damping ratio, spring rate, sprung and un sprung weight, roll Centre, ground clearance, wheel base, track width, caster angle, toe angle, camber angle etc.

Design consideration:

Kingpin and caster angle are kept in such a way that they can compensate each other camber gain.

♣ A positive king pin angle is kept to help in steering the vehicle.

♣ Roll centre below CG to prevent roll over and helps the vehicle to corner better.

♣ Front ride frequency is greater than rear.

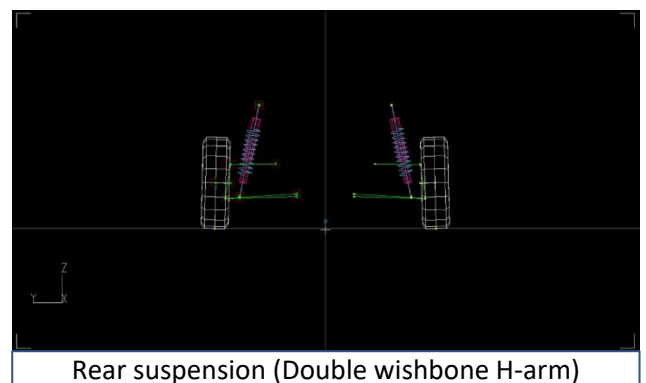
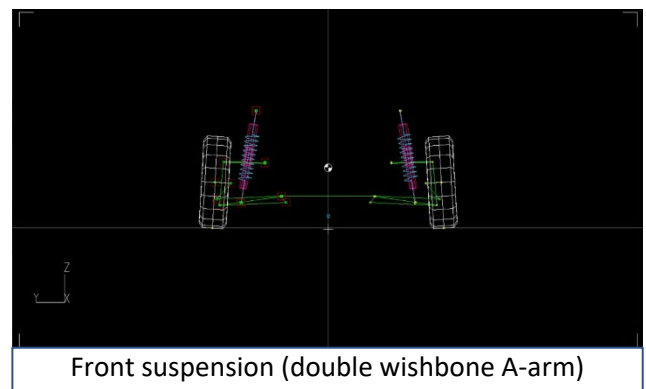
♣ Roll axis inclined towards front to give under-steer characteristic.

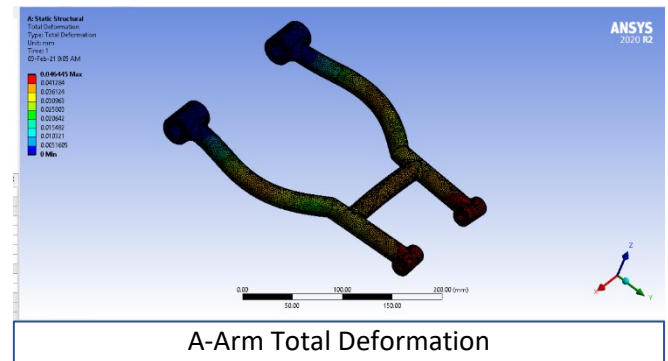
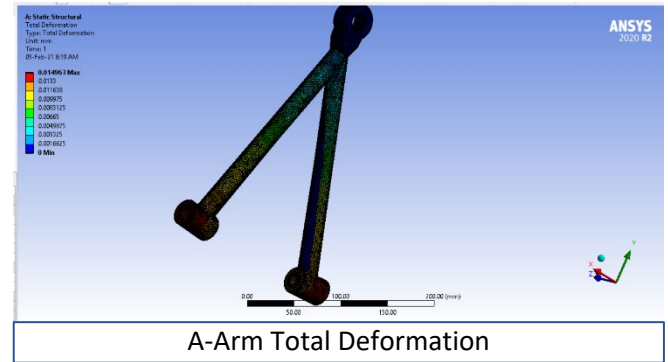
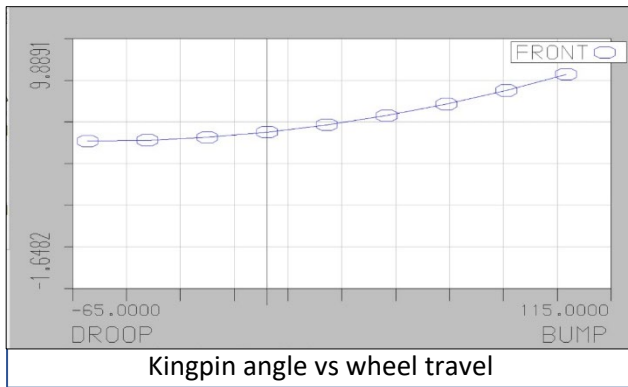
♣ We have taken double wishbone A-arm suspension in the front and double wishbone H-arm as in the case of rear suspension.

We proceeded by using these parameters and by checking various properties such as roll motion, bump motion and steering motion. These properties are controlled by controlling the camber, caster, toe, kingpin angle etc.

Numerical Results: -

- Camber angle : -2.305°
- Caster angle : 0.3°
- King pin Inclination : 7.125°
- Toe in : 0
- Scrub radius : 32.6mm





Calculations:

FOR FRONT SUSPENSION:

According to the amount of load applied on front wheels in different cases maximum applied force was calculated to design the spring. For designing the spring, mainly two cases were taken into account:

- Load transfer on front wheels due to braking
- Load transfer due to dropping the vehicle from a certain height (4G force applied)

Vehicle Dimensions and other Parameters:

Wheel base $b = 1242.4\text{mm}$

Centre of gravity height $h = 400\text{ mm}$

Weight of vehicle with driver $w = 250\text{ kg}$

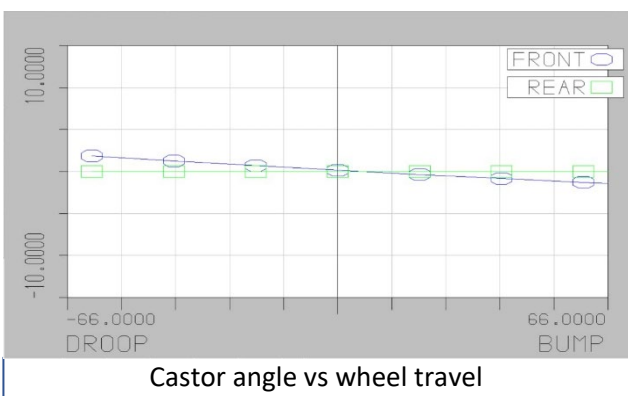
Deceleration produced $a = 1.5\text{ G}$

CASE 1:

Longitudinal Load Transfer on Front Wheels =
 $= w * a * h / b$

$$= 250 * 1.5 * 9.8 * 400 / 1242.4$$

$$= 1183.2\text{ N}$$



Net Reaction at individual wheel = $(1183.2/2) = 591.59 \text{ N}$

Now taking factor of safety as 1.5, net reaction is taken as 887.395 N at individual wheel.

CASE 2:

If load is dropped from a given height taking 4 G force as the reaction, total reaction produced

$$= 4 \times 9.8 \times 250$$

$$= 9800 \text{ N}$$

Weight Distribution front/rear is 45:55

Net reaction at the front of the vehicle

$$= 9800 \times 45/100$$

$$= 4410 \text{ N}$$

Hence reaction at individual wheel = $(4410/2)$

$$= 2205 \text{ N}$$

FOR REAR SUSPENSION:

The net load acting on wheels = 250×9.8

$$= 2450 \text{ N}$$

According to design the centre-to-centre distance of suspension arm is 24 inches

The spring and damper system would be mounted 16 inches from front end.

Thus, considering the forces on the arm in the event of bump, the spring comes out as follows from the moment equation, we get:

$$2450 \times 24 = SF' \times 16$$

$$SF' = 3675 \text{ N}$$

This SF' force is the vertical component of the spring force acting in the upward direction.

Taking the cos component, net force on the spring comes out as

$$SF = SF' / \cos(16)$$

$$= 3823.1 \text{ N}$$

By considering the appropriate suspension travel as 7 inches to accommodate extreme forces, we get:

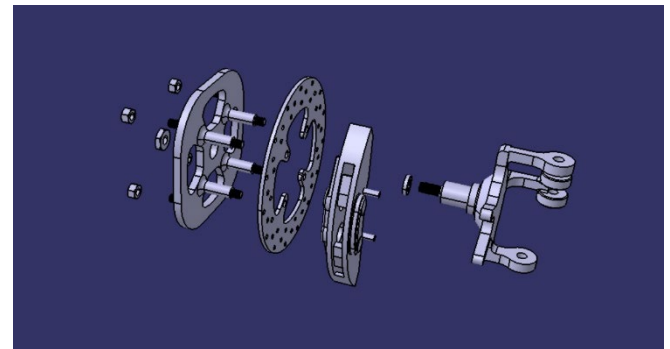
$$\text{Stiffness, } k = 3823.1/7 \times 25.4$$

$$k = 21.5 \text{ N/mm}$$

WHEEL ASSEMBLY

Front wheel assembly

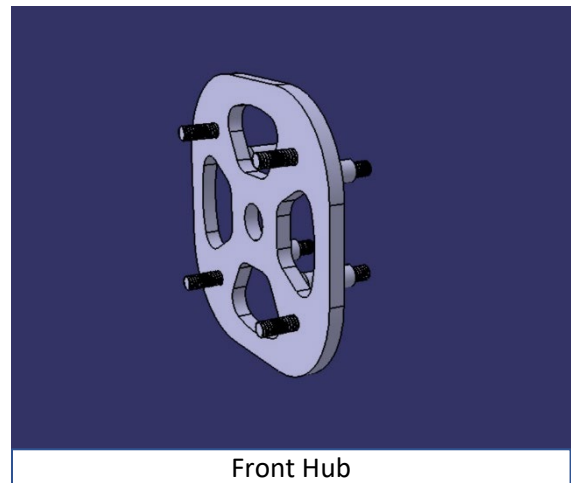
Wheel assembly is designed according to the pre-assumed parameters in steering and suspension. The goal was to have a compact inboard packaging



Front Hub Assembly

Hub Geometry

The wheel hubs are designed as per standard rim size and analytical calculations have been on the basis of shear failure and torsional failure. The material selected is EN 18 due to its extensive strength, light weight, and optimum cost.

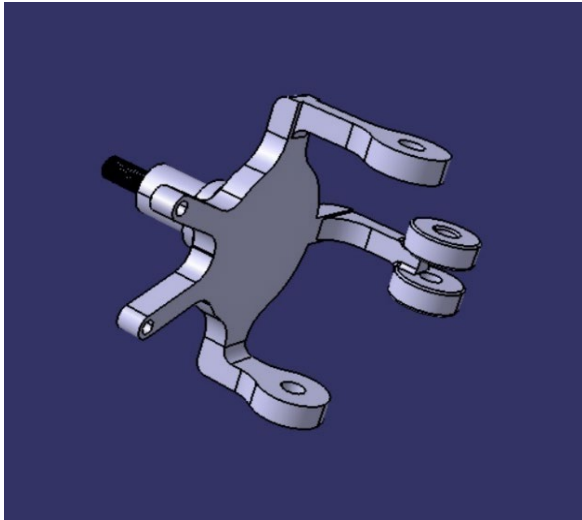


Front Hub

Front Knuckle

Knuckle is designed according to the specification required for 3-Link suspension system and CV joint. Material selected is EN18 due to its good ductility and shock resistant and its resistance to wear properties.

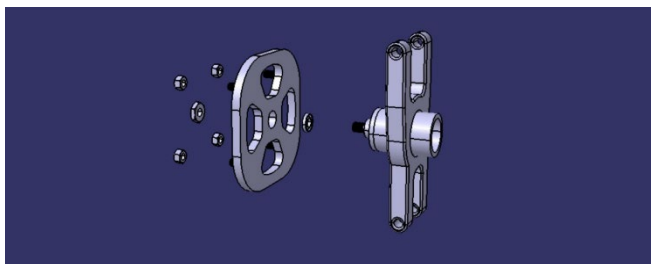
King-Pin Inclination - 7.125°



Front Knuckle

Rear wheel Assembly

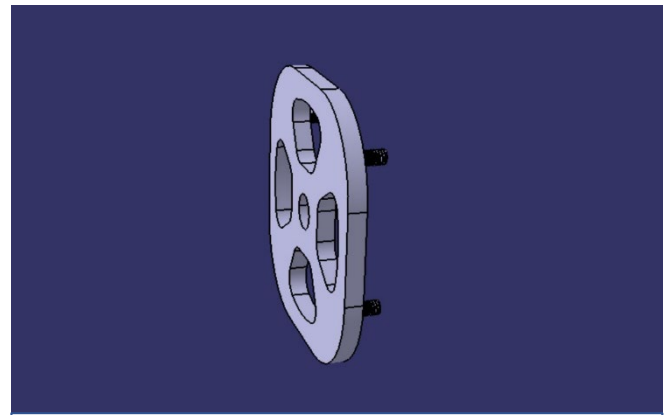
Offset for rear wheel assembly is given by considering length of Wheel side CV joint



Rear Hub Assembly

Hub Geometry

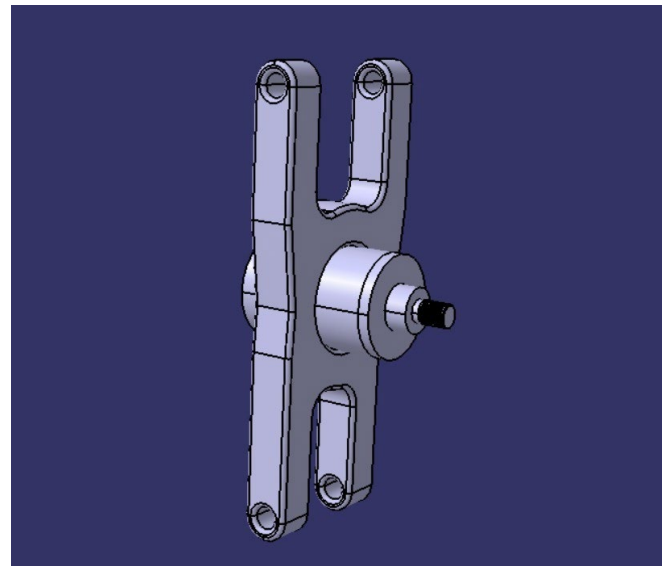
A short circular geometry hub is preferred for rear wheel assembly due its low cost and durability. Hub is designed by considering dimensions of CV joint.



Rear Hub

Rear Knuckle

Knuckle is designed according to the specification required for 3-Link suspension system and CV joint. Material selected is EN19 due to its good ductility and shock resistant and its resistance to wear properties



Rear Knuckle

Wheels and Tire Selection

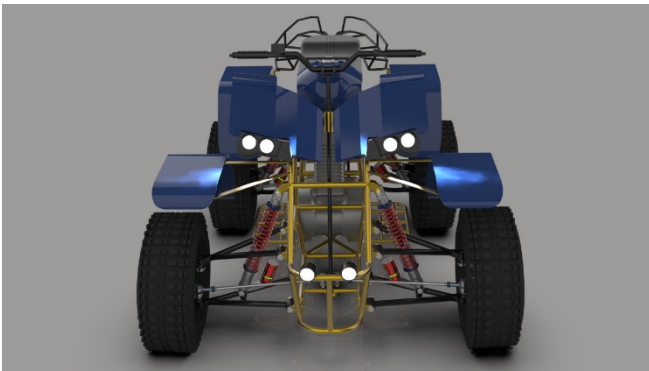
We have chosen 4 ply tires because they have more durability. Deep, thick lugs provide traction in the harshest mud conditions. Off-road trail riding varies from tight single-track trails in the mountains, down to high-speed wide-open desert riding and everything in between Tires.

Tires Dimensions

Front: 24*8*R12

Rear: 24*8*R12

Rendering



REFERENCE

- Rule Book, Quad Torc 2020
- Theory of Machines by S.S. Rattan
- Brake Design and Safety by Rudolf Limpert
- ATV Standards by Gesellschaft
- Webinar conducted by DSIM guidance company
- “Design and analysis of an ATV suspension system” –International Journal of Current Engineering and Technology| Volume 7 | Issue 1 | February 2017 ISSN 2277–4106
- “Thermal Analysis of Disc Brake” - IJIRST –International Journal for Innovative Research in Science & Technology| Volume 3 | Issue 12 | May 2017 ISSN (online): 2349-6010
- Various service manuals of ATVs

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